

THE GEORGE WASHINGTON UNIVERSITY

WASHINGTON, DC

Data Science Program
Department of Data Science
DATS 6501 - Capstone Project - Fall 2026

Post-Training Analysis & Novelty Detection Using Neural Network Self-Organizing Maps (NNSOM)

PHASE 1 REPORT

Baseline Model Training & Latent Feature Extraction

Capstone - Group 5

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1. Week 2 Overview

Week 2 completed the baseline modeling stage of the capstone. The MNIST dataset was prepared using a stratified 70/15/15 train-validation-test split, a LeNet-5 convolutional neural network was trained and evaluated, and 84-dimensional feature embeddings were extracted from the penultimate `fc2` layer for later Self-Organizing Map analysis.

2. Related Work

Convolutional neural networks provide the supervised representation-learning foundation for this project. LeCun et al. LeCun et al. 1998 showed that CNNs can learn effective hierarchical features for handwritten character recognition. Our work does not propose a new classifier architecture; instead, it treats a trained CNN as a source of latent representations whose organization and failures can be studied after training.

Self-Organizing Maps offer a complementary way to inspect high-dimensional structure. Kohonen's original SOM Kohonen 1990 preserves neighborhood relationships on a low-dimensional grid, while Vesanto and Alhoniemi Vesanto and Alhoniemi 2000 showed how trained SOM prototypes can support quantitative clustering. More recent methods integrate neural representation learning directly with SOM organization, including SOM-VAE Fortuin et al. 2019 and Deep Embedded Self-Organizing Maps Forest et al. 2021. In contrast, this capstone keeps the supervised CNN fixed during the initial NNSOM analysis so that SOM-derived diagnostic signals can be evaluated separately from representation learning.

A second relevant line of work concerns novelty and out-of-distribution (OOD) detection. Maximum softmax probability provides a simple post-hoc baseline Hendrycks and Gimpel 2017, while ODIN modifies inference using temperature scaling and input perturbations Liang, Li, and Srikant 2018. Lee et al. Lee et al. 2018 use Mahalanobis distance in deep feature space, Deep SVDD learns a compact one-class representation Ruff et al. 2018, and energy-based scoring provides another strong output-space OOD signal Liu et al. 2020. These studies motivate rigorous comparison because they show that useful novelty information can appear in either classifier outputs or latent features.

The distinguishing hypothesis of this project is therefore not that SOMs are automatically superior to these methods. Rather, we test whether NNSOM quantization and topology applied to fixed CNN embeddings expose structured error regions and provide useful novelty signals. This post-training framing connects SOM representation analysis with classifier failure geography while keeping the contribution empirically testable against established baselines.

3. Methodology

Item	**Week 2 Setup**
Dataset	MNIST, 70,000 grayscale images (28×28)
Split	49,000 train / 10,500 validation / 10,500 test
Model	LeNet-5 CNN
Training	20 epochs, Adam optimizer, cross-entropy loss
Feature layer	`fc2` penultimate layer, 84 dimensions

The best validation checkpoint was used for final test evaluation and feature extraction. The saved `fc2` embeddings from all three splits provide the input for NNSOM in Week 3.

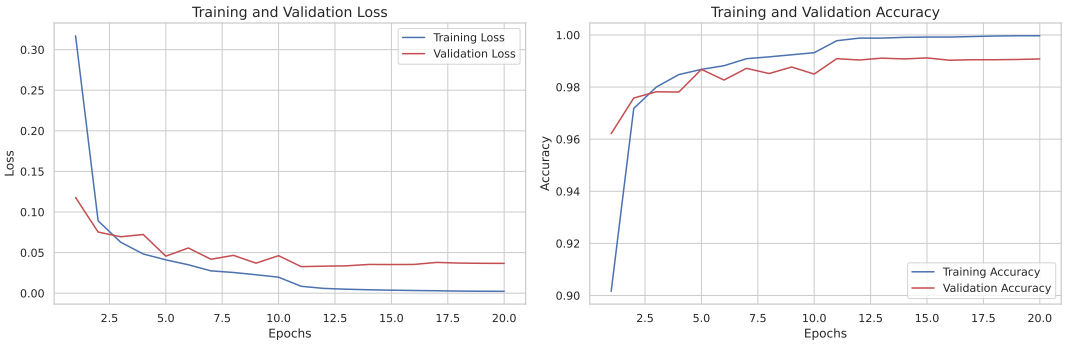


Figure 1. *Training and validation loss/accuracy across 20 epochs.*

4. Results

Metric	**Result**
Test accuracy	99.18% (10,414 / 10,500); 95% Wilson interval 98.99%–99.34%
Macro F1-score	0.9918
Misclassified samples	86 (0.82%)
Embedding output	84-dimensional `fc2` vectors

The baseline generalized well to the held-out test set. The 95% Wilson score interval for the test accuracy, 98.99%–99.34%, shows the uncertainty that comes from evaluating on 10,500 images. Errors were sparse and concentrated among a small number of class pairs; the most frequent individual confusions included $7 \to 1$, $5 \to 8$, and $2 \to 8$, with four cases each. Digit 0 had only one misclassification.

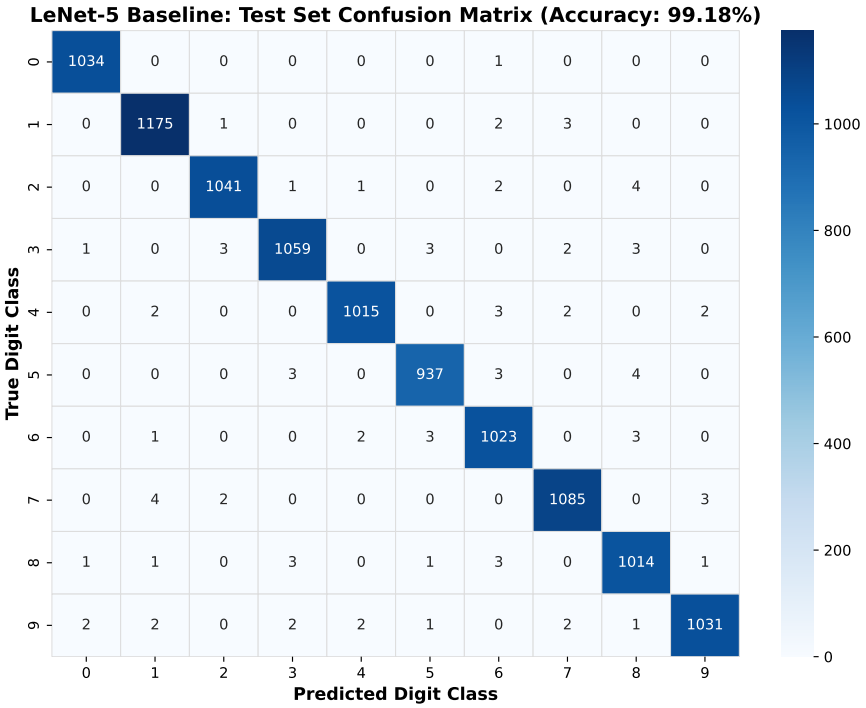


Figure 2. *Test-set confusion matrix. The strong diagonal indicates high class-level accuracy.*

5. Latent Feature Representation

For every sample, the 84-dimensional activation from the `fc2` layer was extracted and saved. This representation captures the learned features immediately before the final classification layer and is the feature space that will be analyzed with NNSOM.

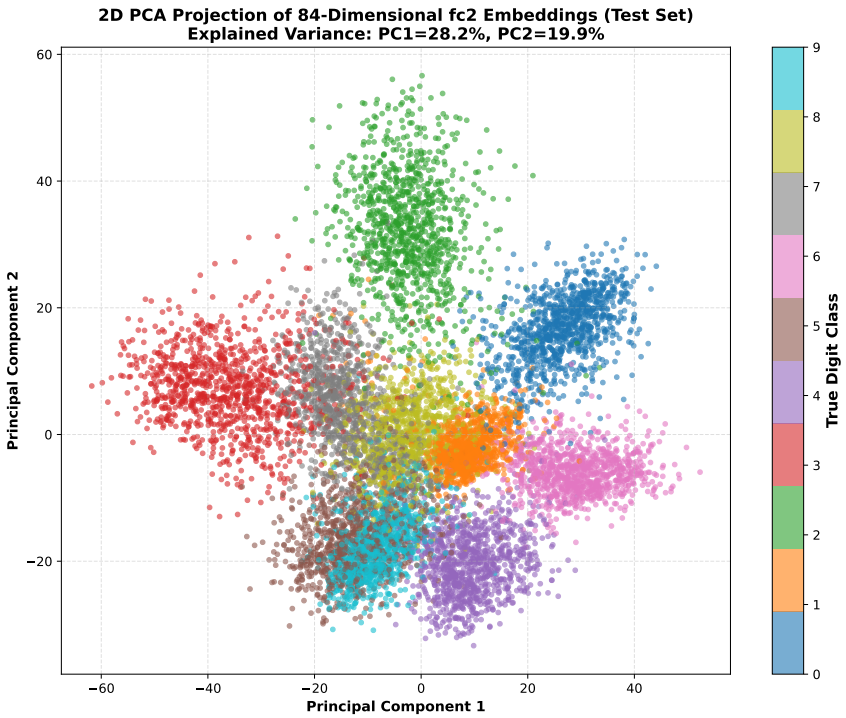


Figure 3. *PCA projection of the 84-dimensional* `fc2` *test embeddings for visualization only.*

The PCA view is only a preliminary 2D visualization; Week 3 will use the full 84-dimensional training embeddings as the actual input to NNSOM.

6. Leakage Control

The test split was kept out of every development decision:

- The 70,000 images were split once, by a stratified split with a fixed seed (42), into 49,000 training, 10,500 validation and 10,500 test images. Every image has a fixed sample ID, so the three splits never overlap.
- Pixels were only scaled to $[0, 1]$ (divided by 255). No statistic was computed from the data for normalization, and no augmentation was used.
- The model was trained on the training split only. The checkpoint was chosen by validation accuracy; the training settings (20 epochs, Adam, learning rate and schedule) were fixed in the configuration file and were not tuned on the test split.
- The test split was evaluated once, after training, for the accuracy, F1-score and confusion matrix reported above.
- Test embeddings were saved for completeness, but the SOM analysis that follows uses the training and validation embeddings only.

7. Conclusion and Week 3 Transition

Week 2 established a reproducible MNIST baseline and produced the feature embeddings required for the next phase. In Week 3, NNSOM Jafari n.d. will be configured and trained on the 49,000 training embeddings to map samples to Best Matching Units, monitor quantization error, and visualize class organization using SOM-based plots such as hit maps and the U-Matrix.

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